

- 0 6 . 1** Give **three** structural differences between an mRNA molecule and a tRNA molecule. **[3 marks]**

| mRNA | tRNA |
|------|------|
|      |      |
|      |      |
|      |      |

- 0 6 . 2** **Table 3** shows mRNA codons and the amino acid coded by each codon.

**Table 3**

| First base | Second base |     |      |             | Third base |
|------------|-------------|-----|------|-------------|------------|
|            | U           | C   | A    | G           |            |
| U          | Phe         | Ser | Tyr  | Cys         | U          |
|            | Leu         |     | Stop | Stop<br>Trp | C          |
| C          | Leu         | Pro | His  | Arg         | A          |
|            |             |     | Gln  |             | G          |
| A          | Ile         | Thr | Asn  | Ser         | U          |
|            | Met         |     | Lys  | Arg         | C          |
| G          | Val         | Ala | Asp  | Gly         | A          |
|            |             |     | Glu  |             | G          |

**Figure 5** shows the mRNA base sequence produced when part of a gene coding for an enzyme is transcribed.

**Figure 5**

|     |     |     |
|-----|-----|-----|
| UUU | CGG | GCG |
|-----|-----|-----|

Use **Table 3** to give the amino acids coded by the mRNA base sequence in **Figure 5**. **[1 mark]**

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

0 6 . 3

A mutation occurred in the part of the gene transcribed in Question 06.2.

**Figure 6** shows the mRNA base sequence produced when the identical part in the mutated gene is transcribed.

**Figure 6**

|     |     |     |
|-----|-----|-----|
| UUU | AGG | GCG |
|-----|-----|-----|

The amino acids coded by this mRNA base sequence form part of the enzyme's active site.

Use all the information in this question to:

- name the type of mutation that occurred to produce the mutated gene
- give the change in **DNA** caused by this mutation
- explain the effect this mutation will have on the function of the enzyme.

**[4 marks]**

Type of mutation \_\_\_\_\_

Change in DNA \_\_\_\_\_

Explanation of effect on function of enzyme \_\_\_\_\_

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**Turn over for the next question**

**Turn over ►**

